

Battery Energy Management (BEM) System

Week 3 Enhanced Deliverable: SADT/IDEF0 Functional Decomposition

FOCUS: Communication Interfaces, Operating Mode Control, OBD Diagnostics

Group 4: Communication & System Integration

Module: Automotive System Architecture: System-level Design

Target: 1st Year Engineering Students in Automotive Software Engineering (ASE1.1)

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Academic Year: 2025-2026

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1. Executive Summary - Addressing Teacher Feedback

This enhanced deliverable addresses the specific feedback provided by the teachers regarding the initial Week 3 submission. The primary criticism was that the original diagrams focused too heavily on the technical and functional aspects of the BMS, while neglecting three critical areas: (1) interfacing and communication specifications, (2) operating mode control logic, and (3) OBD diagnostic services integration. This revision fundamentally restructures the SADT decomposition to place these interface-centric concerns at the forefront of the design.

1.1 Teacher Feedback Analysis

The original submission presented the A4 function primarily from a functional decomposition perspective, describing what the system does rather than how it interfaces with other systems. While technically correct, this approach missed the essential purpose of Group 4's role: to serve as the communication hub and coordination center for the entire BEM system. The following table summarizes the feedback points and how this revised document addresses each one:

Feedback Point	Issue	Enhancement
Interfacing Focus	Original diagrams showed internal functions but not interface details	Added detailed CAN message flow, signal definitions, timing specs
Communication	CAN protocol details were superficial	Full CAN ID mapping, message timing tables, protocol compliance
Operating Modes	Mode control was described but not detailed	Complete state machine with transitions, conditions, ISO 26262 compliance
OBD Diagnostics	Diagnostic services were mentioned but not elaborated	Full UDS service implementation, DID mapping, protocol stack

Table 1: Teacher Feedback and Enhancement Mapping

1.2 Interface-Centric Design Philosophy

The revised SADT diagrams now follow an interface-centric design philosophy where each functional block is primarily characterized by its external interfaces rather than its internal operations. This approach aligns with automotive industry best practices, where interface control documents (ICDs) are critical artifacts for system integration. The ICOM (Inputs, Controls, Outputs, Mechanisms) model has been extended to include specific protocol requirements, timing constraints, and signal-level definitions that enable direct implementation in MATLAB/Simulink or other model-based development tools.

2. A4 Enhanced Decomposition - Interface-Oriented Design

The enhanced A4 decomposition restructures the sub-functions around their interface responsibilities rather than their internal operations. Each sub-function is now characterized primarily by the signals it exchanges with other systems, the protocols it implements, and the timing constraints it must satisfy. This interface-centric view provides a clearer foundation for system integration and verification activities.

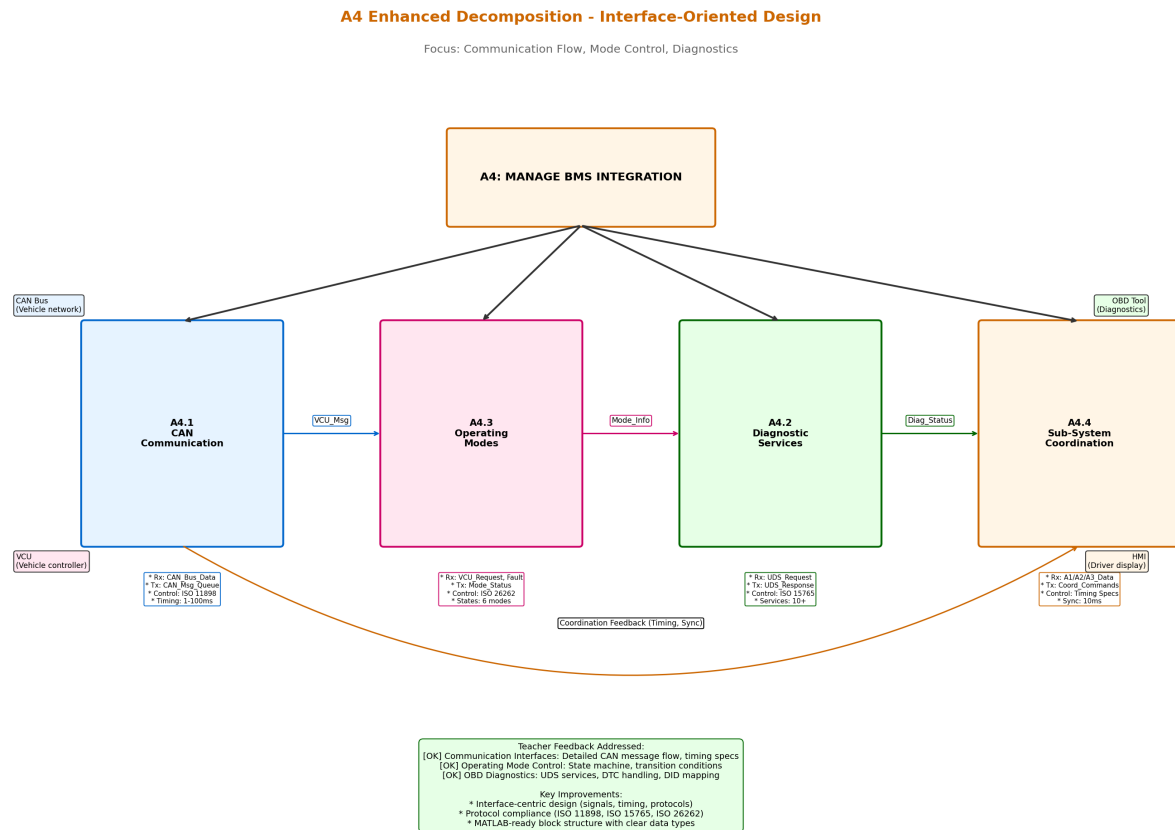


Figure 1: A4 Enhanced Decomposition with Interface Focus

The decomposition shows four sub-functions arranged to emphasize their interface responsibilities: A4.1 handles all CAN communication with the vehicle network; A4.3 manages the BMS operating mode state machine; A4.2 implements diagnostic services for service and troubleshooting; and A4.4 coordinates data flow between internal BMS functions. The arrows between functions now represent specific signal flows with defined timing requirements, making this diagram directly implementable in MATLAB/Simulink.

3. Communication Interface Architecture (A4.1)

The A4.1 function "Manage CAN Communication" is the primary interface between the BMS and the vehicle network. This function handles all message transmission and reception on the CAN bus, implementing the ISO 11898 (CAN 2.0B) protocol with specific timing and priority requirements derived from the vehicle-level communication matrix.

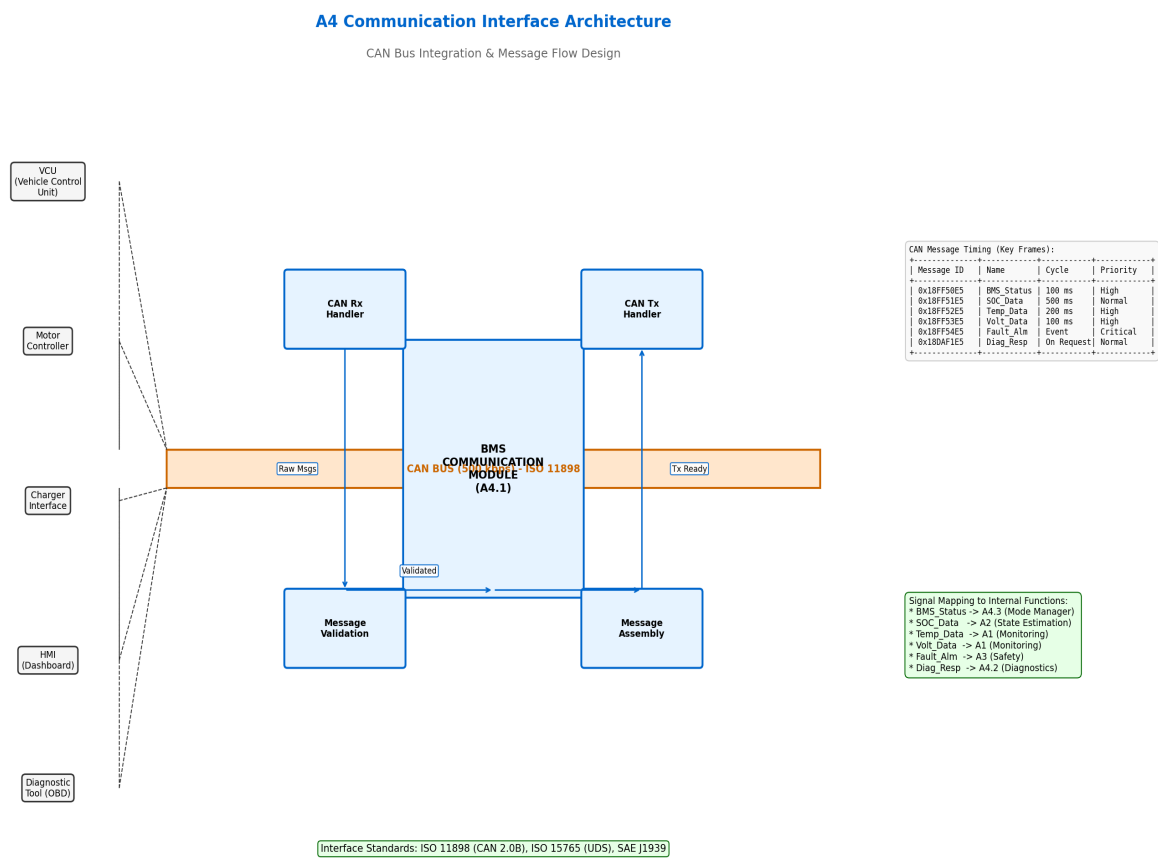


Figure 2: A4.1 Communication Interface Architecture

3.1 CAN Message Timing Specifications

The CAN communication follows a deterministic schedule based on the SAE J1939 extended addressing format. Each message has defined cycle times, priorities, and data content. The BMS transmits periodic status messages at 100ms intervals, while diagnostic responses are transmitted on request. Critical fault messages are transmitted immediately (event-driven) with the highest priority to ensure rapid fault notification to the vehicle control unit.

Message ID	Name	Cycle Time	DLC	Priority	Destination
0x18FF50E5	BMS_Status	100 ms	8	High (6)	VCU, HMI
0x18FF51E5	SOC_Data	500 ms	8	Normal (4)	VCU, HMI
0x18FF52E5	Temperature_Data	200 ms	8	High (6)	VCU
0x18FF53E5	Voltage_Data	100 ms	8	High (6)	VCU
0x18FF54E5	Fault_Alarm	Event	8	Critical (8)	VCU, HMI
0x18DAF1E5	Diag_Response	On Request	Variable	Normal (4)	OBD Tool

Table 2: CAN Message Definition Table

3.2 Signal Mapping to Internal Functions

Each CAN message carries signals that must be routed to the appropriate internal BMS functions. The BMS_Status message contains mode information for A4.3 (Mode Manager), SOC and SOH data from A2 (State Estimation), and overall system health indicators. Temperature and voltage data are forwarded to A1 (Monitoring) for analysis, while fault alarms trigger responses in A3 (Safety). This signal routing is implemented in the Message Validation function within A4.1.

4. Operating Mode State Machine (A4.3)

The A4.3 function "Manage Operating Modes" implements the BMS state machine that controls the overall operational behavior of the battery management system. This state machine is designed to comply with ISO 26262 functional safety requirements, ensuring that the BMS can transition to a safe state from any operating mode within the fault tolerance time interval (FTTI).

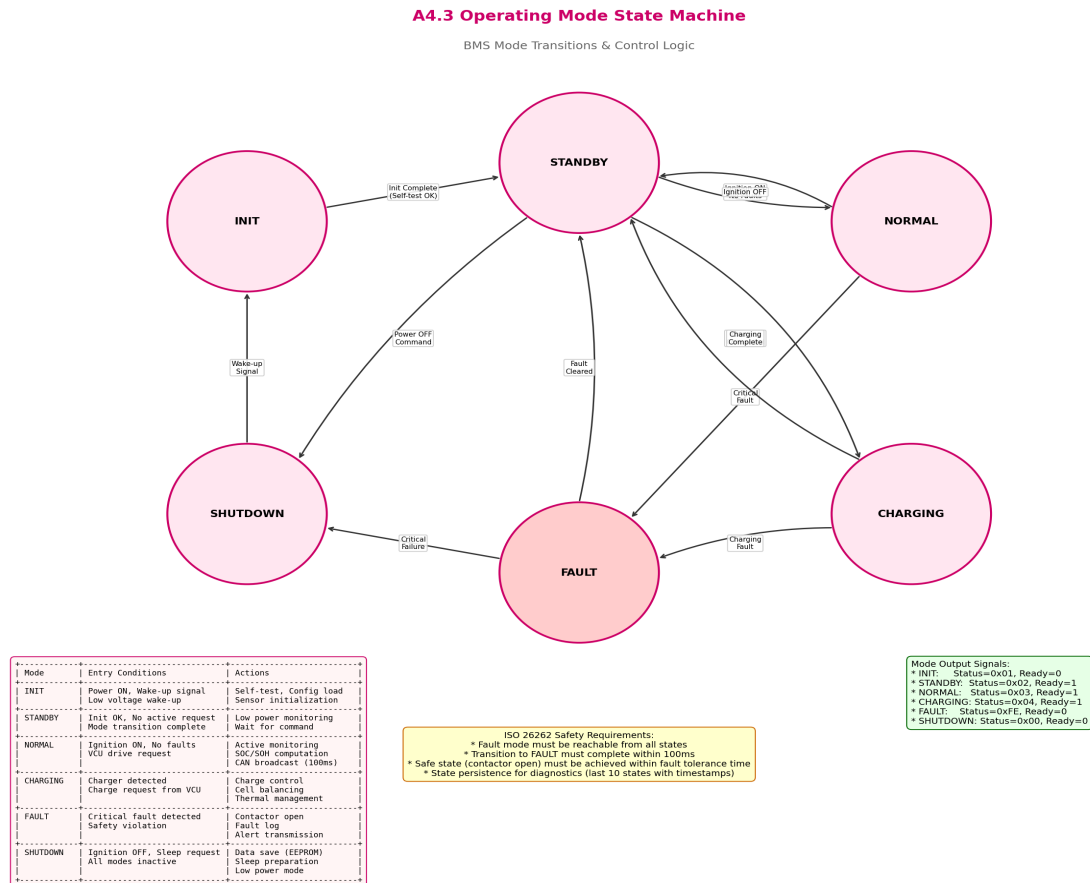


Figure 3: A4.3 Operating Mode State Machine

4.1 Mode Definitions and Entry Conditions

The BMS operates in six distinct modes, each with specific entry conditions, operational behaviors, and exit transitions. The INIT mode performs self-test and configuration loading upon power-on or wake-up. STANDBY mode represents the ready state where the BMS monitors the battery while waiting for vehicle commands. NORMAL mode is active during driving, with full monitoring and state estimation activities. CHARGING mode handles charge control and cell balancing. FAULT mode represents the safe state where contactors are open and the system awaits fault clearance. SHUTDOWN mode prepares the system for low-power sleep.

Mode	Entry Conditions	Actions	Status Code
INIT	Power ON, Wake-up signal	Self-test, Config load, Sensor init	0x01
STANDBY	Init complete, No active request	Low power monitoring, Wait for command	0x02
NORMAL	Ignition ON, No faults, VCU drive request	Active monitoring, SOC/SOH computation, CAN broadcast (100ms)	0x03
CHARGING	Charger detected, Charge request from VCU	Charge control, Cell balancing, Thermal management	0x04

FAULT	Critical fault detected, Safety violation	Contactors open, Fault log, Alert transmission	0xFE
SHUTDOWN	Ignition OFF, Sleep request	Data save (EEPROM), Sleep preparation, Low power mode	0x00

Table 3: BMS Operating Mode Definitions

4.2 ISO 26262 Safety Requirements for Mode Control

The operating mode state machine is designed to meet ISO 26262 ASIL C requirements for functional safety. Key safety requirements include: (1) The FAULT mode must be reachable from all other modes within the fault tolerance time interval of 100ms; (2) All mode transitions must be deterministic and verifiable; (3) The safe state (FAULT mode with contactors open) must be achieved without requiring external inputs; (4) A history of the last 10 mode transitions with timestamps must be maintained for diagnostic purposes; and (5) Mode persistence across power cycles must ensure graceful recovery from unexpected shutdowns.

5. OBD-II/UDS Diagnostic Services (A4.2)

The A4.2 function "Manage Diagnostic Services" implements the diagnostic communication interface that enables service technicians to access BMS data, retrieve diagnostic trouble codes (DTCs), and perform troubleshooting. This function implements the Unified Diagnostic Services (UDS) protocol defined in ISO 15765-3 (ISO 14229) over CAN, as well as OBD-II requirements for emissions-related diagnostics defined in SAE J1979.

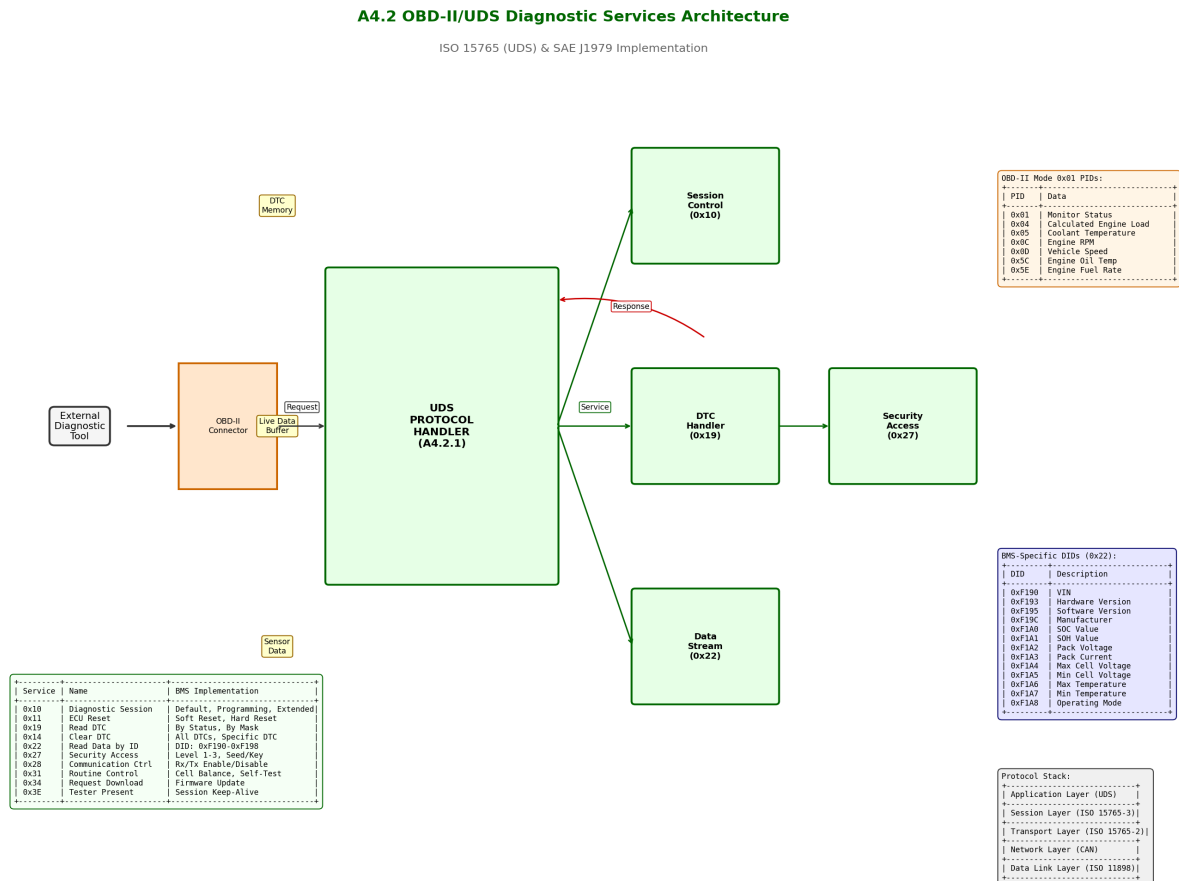


Figure 4: A4.2 OBD-II/UDS Diagnostic Services Architecture

5.1 UDS Service Implementation

The BMS implements a comprehensive set of UDS services tailored to battery management system diagnostics. The Session Control service (0x10) manages diagnostic sessions including Default, Programming, and Extended sessions. The DTC Handler (0x19) reads diagnostic trouble codes with various filtering options. The Data Stream service (0x22) reads data by identifier, providing access to battery-specific parameters. Security Access (0x27) protects critical functions with a seed-key authentication mechanism. Additional services include ECU Reset (0x11), Communication Control (0x28), Routine Control (0x31) for cell balancing and self-test, and Tester Present (0x3E) for session keep-alive.

5.2 BMS-Specific Data Identifiers (DIDs)

The BMS defines a set of BMS-specific Data Identifiers (DIDs) for accessing battery parameters through the UDS ReadDataByIdentifier service (0x22). These DIDs follow the ISO 14229 convention where 0xF1xx identifiers are manufacturer-specific. The table below lists the key DIDs implemented in the BMS diagnostic interface:

DID	Description	Data Type	Access
0xF190	Vehicle Identification Number (VIN)	17-byte ASCII	Read

0xF193	Hardware Version	4-byte string	Read
0xF195	Software Version	4-byte string	Read
0xF1A0	State of Charge (SOC)	uint8 (percentage)	Read
0xF1A1	State of Health (SOH)	uint8 (percentage)	Read
0xF1A2	Pack Voltage	uint16 (0.1V resolution)	Read
0xF1A3	Pack Current	int16 (0.1A resolution)	Read
0xF1A4	Maximum Cell Voltage	uint16 (1mV resolution)	Read
0xF1A5	Minimum Cell Voltage	uint16 (1mV resolution)	Read
0xF1A6	Maximum Temperature	int8 (deg C offset 40)	Read
0xF1A7	Minimum Temperature	int8 (deg C offset 40)	Read
0xF1A8	Operating Mode	enum (0-5)	Read

Table 4: BMS-Specific Data Identifiers

6. Cross-Team Interface Control Document

The Interface Control Document (ICD) defines the precise signal-level interfaces between Group 4 (A4) and the other three teams, as well as external vehicle systems. This document is critical for system integration, ensuring that all teams develop compatible components with consistent data formats, timing requirements, and safety constraints.

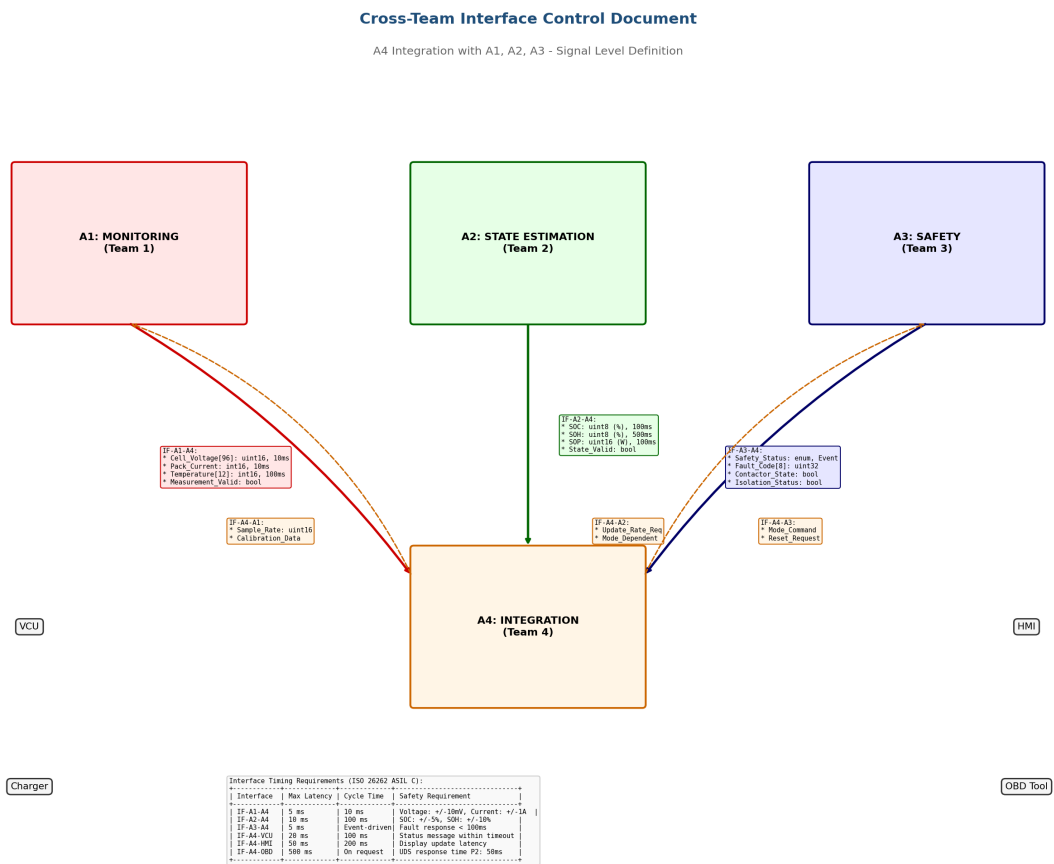


Figure 5: Cross-Team Interface Control Diagram

6.1 Signal-Level Interface Definitions

Each interface is defined at the signal level with data types, update rates, and direction. The interface between A4 and A1 (Monitoring) carries measurement data: Cell_Voltage[96] as uint16 array at 10ms cycle time, Pack_Current as int16 at 10ms cycle time, and Temperature[12] as int16 array at 100ms cycle time. The A4-A2 interface carries state estimation data: SOC (uint8, 100ms), SOH (uint8, 500ms), and SOP (uint16, 100ms). The A4-A3 interface carries safety-critical data: Safety_Status (enum, event-driven), Fault_Code[8] (uint32 array), and Contactors_State (bool).

6.2 Timing and Safety Requirements

All interfaces must satisfy timing requirements derived from ISO 26262 ASIL C safety goals. The measurement interface (A4-A1) has a maximum latency of 5ms to support real-time monitoring. The state estimation interface (A4-A2) tolerates 10ms latency due to the slower dynamics of SOC/SOH changes. The safety interface (A4-A3) has the most stringent requirement of 5ms maximum latency with event-driven updates for immediate fault notification. External interfaces to the VCU must complete status message transmission within the 100ms cycle time, while OBD diagnostic responses must satisfy the UDS P2 timing requirement of 50ms.

7. MATLAB/Simulink Implementation Guide

The functional decomposition and interface definitions in this document are designed to be directly implementable in MATLAB/Simulink for model-based development. The SADT diagrams translate to Simulink subsystems with defined input/output ports, data types, and timing requirements that support automatic code generation for embedded targets.

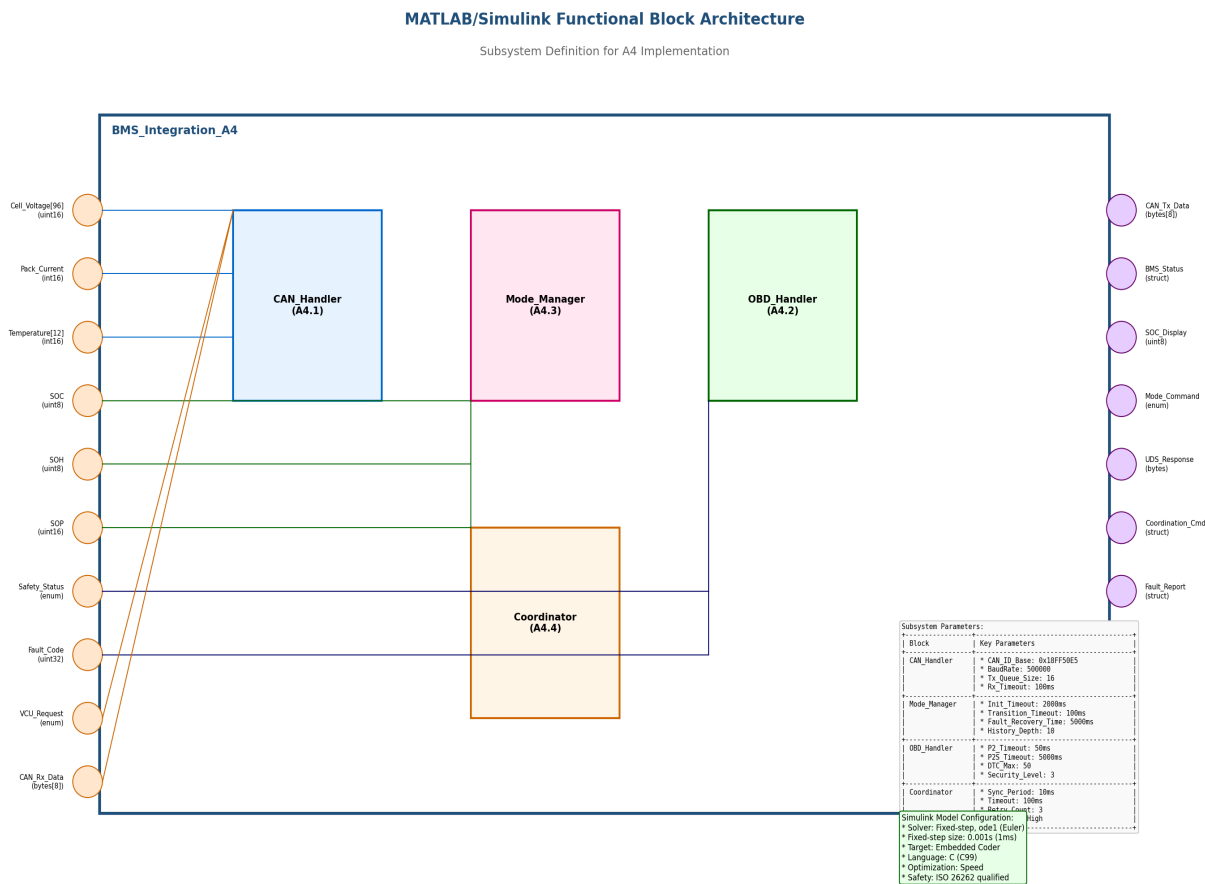


Figure 6: MATLAB/Simulink Functional Block Architecture

7.1 Subsystem Architecture

The main A4 subsystem (BMS_Integration_A4) contains four major subsystems: CAN_Handler, Mode_Manager, OBD_Handler, and Coordinator. Each subsystem has well-defined input and output ports with specific data types matching the interface definitions in the ICD. The CAN_Handler subsystem implements the CAN message processing logic with message ID filtering, signal extraction, and message assembly. The Mode_Manager subsystem implements the state machine using Simulink Stateflow for clear state transition modeling. The OBD_Handler subsystem implements the UDS protocol handler with service dispatching and response assembly. The Coordinator subsystem manages data synchronization between the other subsystems.

7.2 Model Configuration for Code Generation

The Simulink model is configured for embedded code generation targeting the BMS microcontroller. Key configuration settings include: Fixed-step solver with ode1 (Euler) method for deterministic timing, step size of 0.001s (1ms) matching the fastest sampling requirement, Embedded Coder target with C99 language compliance, speed optimization, and ISO 26262 qualification for safety-critical code generation. The generated code integrates with the CAN driver stack and diagnostic services implementation on the target hardware.

8. Requirement Traceability Matrix Update

The Requirement Traceability Matrix (RTM) has been updated to reflect the enhanced interface-focused design. Each requirement now traces to specific functional blocks and interface signals, ensuring complete coverage of stakeholder needs through the detailed design.

Req ID	Requirement	Satisfied By	Interface
FR-COM-01	BMS shall communicate via CAN at 500kbps with defined message IDs	A4.1 (CAN_Handler)	IF-A4-VCU, IF-A4-HMI
FR-COM-02	BMS shall respond to UDS diagnostic requests within 50ms	A4.2 (OBD_Handler)	IF-A4-OBDD
FR-COM-03	BMS shall broadcast status every 100ms during normal operation	A4.1 (CAN_Handler)	IF-A4-VCU
FR-MOD-01	BMS shall implement operating modes per ISO 26262	A4.3 (Mode_Manager)	Internal
FR-MOD-02	BMS shall transition to FAULT mode within 100ms of critical fault	A4.3 (Mode_Manager)	IF-A3-A4
FR-DIAG-01	BMS shall provide DTC storage and retrieval via UDS	A4.2 (OBD_Handler)	IF-A4-OBDD
FR-DIAG-02	BMS shall provide live data via DID 0xF1A0-0xF1A8	A4.2 (OBD_Handler)	IF-A4-OBDD
NFR-TIM-01	All interfaces shall meet defined latency requirements	A4.4 (Coordinator)	All
NFR-SAF-01	All communication shall comply with ISO 26262 ASIL C	A4.1, A4.3	IF-A3-A4, IF-A4-VCU

Table 5: Enhanced Requirement Traceability Matrix

9. Conclusions and Next Steps

This enhanced deliverable addresses the teacher feedback by fundamentally restructuring the SADT decomposition around interface-centric design principles. The diagrams now clearly show the communication interfaces, operating mode control logic, and OBD diagnostic services that are central to Group 4's role as the integration hub of the BMS system. The detailed signal definitions, timing requirements, and protocol specifications provide a solid foundation for implementation in MATLAB/Simulink.

9.1 Key Improvements Made

The following improvements have been implemented: (1) Communication interfaces are now defined at the signal level with CAN message IDs, timing, and protocol compliance details; (2) Operating mode control is represented as a complete state machine with transition conditions and ISO 26262 safety requirements; (3) OBD diagnostic services include full UDS service implementation and BMS-specific DID definitions; (4) Interface Control Document provides signal-level definitions for all cross-team interfaces; (5) MATLAB/Simulink implementation guide shows how the SADT decomposition maps to model-based development artifacts.

9.2 Next Steps for Week 4

Building on this enhanced functional architecture, Week 4 activities will focus on: (1) Simulink model development for each subsystem with Stateflow state machines; (2) CAN message database (DBC file) creation based on the message definitions; (3) UDS protocol stack integration for diagnostic services; (4) Hardware-in-the-loop (HIL) test case definition for interface verification; and (5) Safety analysis refinement per ISO 26262 Part 4 for system design.